Insights from Open Source Student Engagement Efforts at Georgia Tech

Dr. Jeffrey Young, Alex Jenkins

Apereo Microconference, July 2026
July 15th, 2026

Georgia Tech

How Did GT Start an Open Source Program Office (OSPO)?

Georgia Tech received a generous grant (and renewal) from Sloan Foundation in 2023 as part of a push to create new academic OSPOs.

New Office to Provide Open-Source Expertise for Campus Community

December 1, 2023 Uncategorized

A philanthropic award is transforming the future of open-source software at Georgia Tech and establishing a new resource for students and faculty.

A recent $623,790 grant from the Alfred P. Sloan Foundation is funding the creation of the Open Source Program Office (OSPO). The office will provide expert advice and the latest information to students and faculty interested in open-source research and publishing. The grant will support the office in its

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OSPO@GT Team Members

Jeff Young
PI, Director

Fang (Cherry) Liu
Co-PI, Associate Director

Ron Rahaman
Senior Personnel

GT's OSPO is a joint effort including the College of Computing, OIT/PACE and Library!

Cliff Landis
Digital Curation Archivist

Dillon Henry
Digital Accessioning Archivist

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Core OSPO@GT Goals

Improve OSS Developer Engagement

Create a welcoming community for all developers and engage in robust community outreach

Develop added metrics to track OSPO progress and criteria for success

Support Developers at GT and other Atlanta universities through workshops, training sessions, and meet-ups

Institutionalize Student Programs

Evolve the virtual summer internship program to be more sustainable and allow for more opportunity for all interested students

Enhance and Improve training materials to allow for more large-scale, asynchronous OSS training programs

Address the AI Revolution

Educate students on the benefits and pitfalls of open-source AI and current viewpoints, including the ethical use of AI

Host Open Data Sets and exploration tools for AI data agents

Provide training for AI hardware and software tools on local HPC resources

Design joint software engineering training with other GT groups

Explore GT Strengths in OSS

Collaborate with Georgia Tech's School of Cybersecurity and GTRI to develop and share secure open-source best practices.

Support institute efforts in maintaining research integrity using open-source techniques

Core OSPO@GT Goals – Student Focused Efforts!

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Highlighted areas specifically focus on students at GT and beyond!

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How Does OSPO@GT Support Students?

Training & Education

Hands-on workshops and training on open-source skills—from Git, testing, and CI/CD to security, governance, and OSS project lifecycles.

Funding Support

Financial backing for student open-source work, including the paid Virtual Summer Internship Program and support for participation in OSS projects.

OSS Community Building

Industry sponsored events to support open-source topics. Poster sessions to share OSS software on GT’s campus. Support for student-run clubs

Open-Source AI Resources

Access to open compute, model hosting, and infrastructure to help students experiment with and deploy open-source AI.

OSPO@GT Student Impacts

Training & Education

Hands-on workshops and training on open-source skills—from Git, testing, and CI/CD to security, governance, and OSS project lifecycles.

~30 classes, workshops, and meetups taught over the last 3 years

Funding Support

Financial backing for student open-source work, including the paid Virtual Summer Internship Program and support for participation in OSS projects.

Financial Support for 80+ students via VSIP and student sponsorship

OSS Community Building

Industry sponsored events to support open-source topics. Poster sessions to share OSS software on GT's campus. Support for student-run clubs

5 large community events with focus on community engagement since 2023

Open-Source AI Resources

Access to open compute, model hosting, and infrastructure to help students experiment with and deploy open-source AI.

Growing support for open-source AI on GT compute resources

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OSPO@GT – Training & Education

OSS Training

- Growing corpus of open source and software engineering related training
- Supports our Virtual Summer Internship Program

Ongoing Work

- Asynchronous OSS certificate programs
- Future work – classes for undergrads and grads specifically

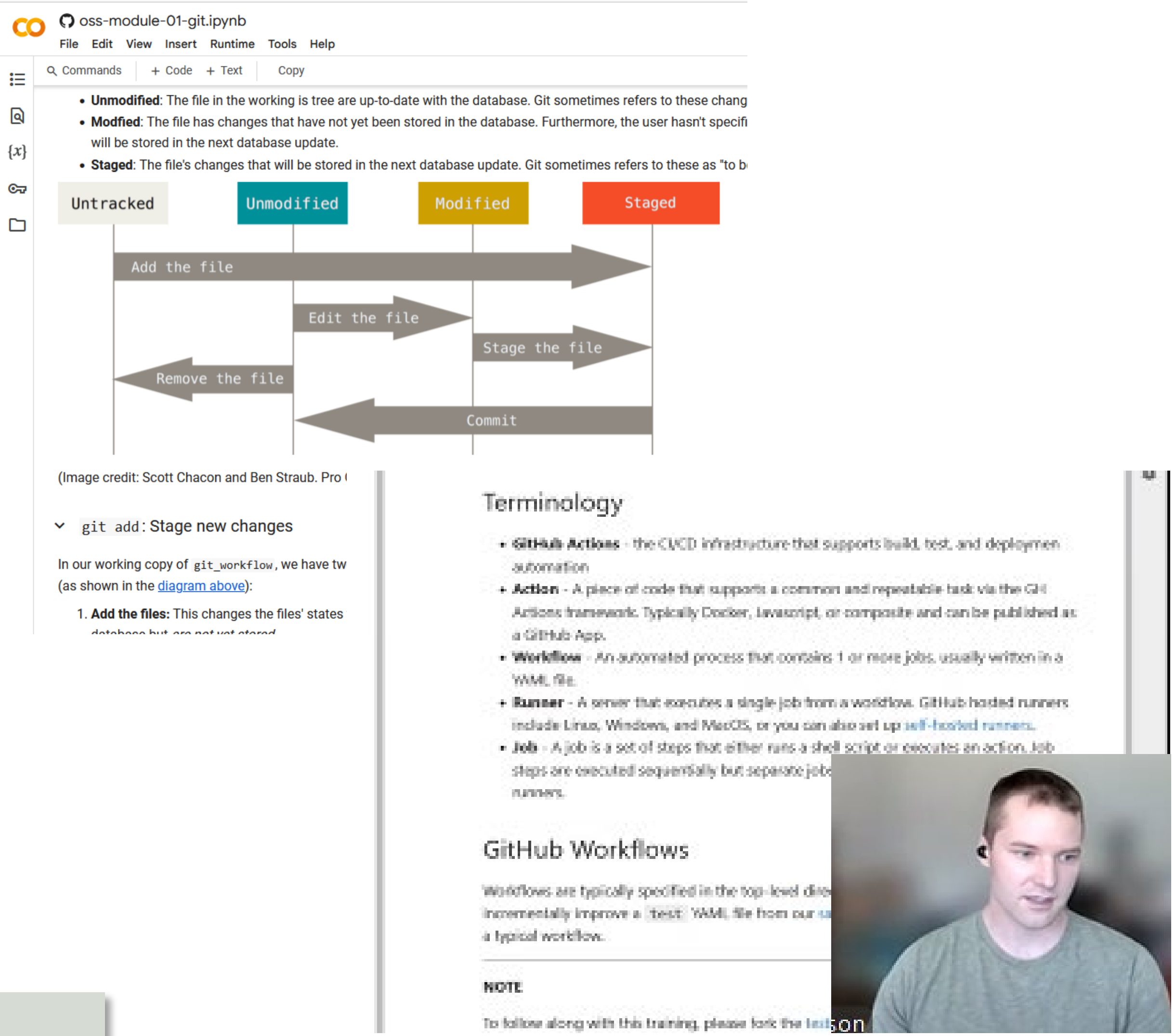
Training & Education

Hands-on workshops and training on open-source skills—from Git, testing, and CI/CD to security, governance, and OSS project lifecycles.

OSPO@GT - OSS Training

8 lessons on open-source training topics developed with PACE and CSSE

- Git – forks, issues, PRs
- Testing Code
- CI/CD
- Reviewing Code
- OSS Governance Models
- OSS Project Lifecycles



Training Lecture on Testing by Daniel Bartilinson

This curriculum recognizes that software engineering (SE) best practices are critical for the sustainability of open source ecosystems

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Learn more at https://ospo.cc.gatech.edu/oss-training/

Ongoing Work - Asynchronous Training Capabilities

Key challenge:

GT has a large study body; how can we teach all of our students key open source concepts?

Asynchronous training certificate program in OSS basics

- Focusing on mapping existing OSS training to open source **nbgrader** framework
- Currently working on Git basics but anticipate added OSS classes, software engineering

Before you turn this problem in, make sure everything runs as expected. First, **restart the kernel** (in the menubar, select Kernel → Restart) and then **run all cells** (in the menubar, select Cell -

Make sure you fill in any place that says `YOUR CODE HERE` or "YOUR ANSWER HERE", as well as your name and collaborators below:

```
[ ]: NAME = ""
     COLLABORATORS = ""
```

Checkout commit that goes back to Lesson 2 start point

```
[1]: import os
     from git_state_manager import save_git_state, load_git_state
     os.chdir("oss-training-templates")

[2]: def checkout_commit():
         # YOUR CODE HERE
         command = !git checkout d4034cff631ebda366cb68353368ddde7eb25c32

         return command

     print(checkout_commit())
     print("------------------------------------------------------------------\n" + save_git_state("git_state_2_2.json", "test_1"))

["Note: switching to 'd4034cff631ebda366cb68353368ddde7eb25c32'.", '', "You are in 'detached HEAD' state. You can look around, make experimental", 'changes and com
n discard any commits you make in this', 'state without impacting any branches by switching back to a branch.', '', 'If you want to create a new branch to retain c
ou may', 'do so (now or later) by using -c with the switch command. Example:', '', '  git switch -c <new-branch-name>', '', 'Or undo this operation with:', '', '
Turn off this advice by setting config variable advice.detachedHead to false', '', 'HEAD is now at d4034cf First commit of my_abs (no try/except yet)']
------------------------------------------------------------------
Saved git state for test_1.

[3]: git_state = load_git_state("git_state_2_2.json", "test_1")

     expected_commit = "d4034cff631ebda366cb68353368ddde7eb25c32"
     assert git_state["commit"] == expected_commit, f"Expected commit '{expected_commit}', found '{git_state['commit']}'"

[4]: def checkout_main():
         # YOUR CODE HERE
         command = !git checkout main

         return command
```

Ongoing Work – Combining Training and Experiences

We'd like to build on our own training and community software engineering efforts like NSF INTERSECT to create:

- An undergraduate course that teaches OSS and SE basics and pairs students with projects
    - Existing related efforts at RPI (RCOS), Delaware and other institutions provide a great template for this type of course.
- Opportunities for training and engagement to our cohort of 10,000+ online masters students!

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OSPO@GT – Funding Support

Funding Support

Financial backing for student open-source work, including the paid Virtual Summer Internship Program and support for participation in OSS projects.

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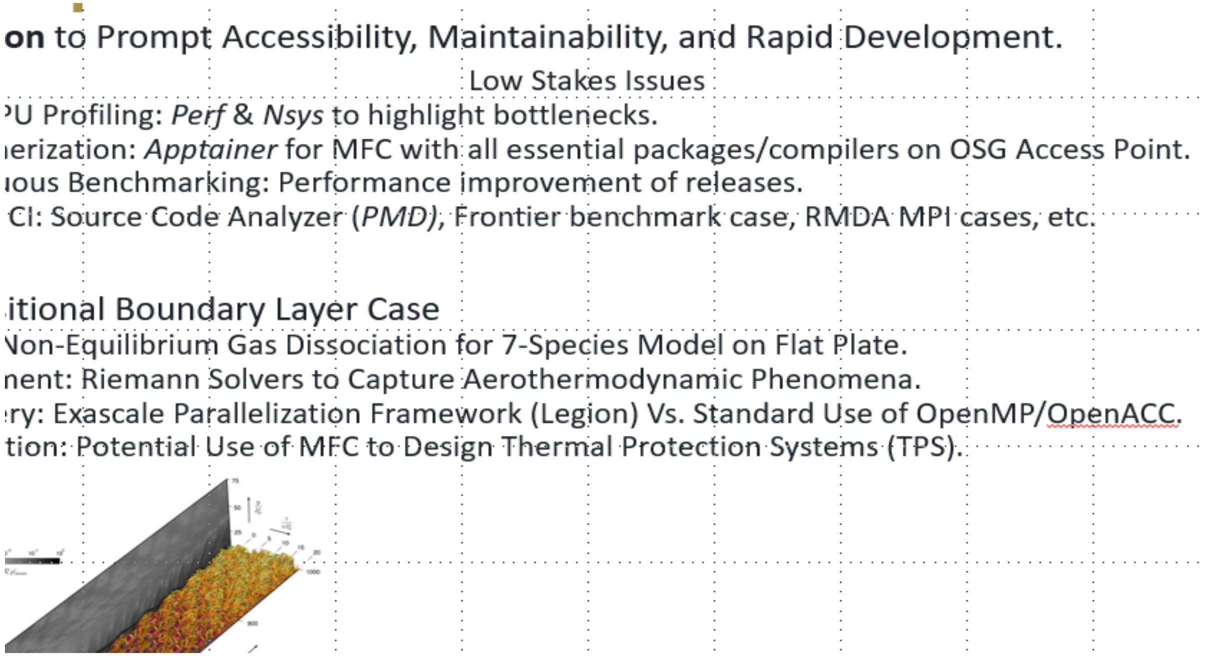
OSPO@GT - Virtual Summer Internship (Summer 2024-2026)

Virtual summer internship program (VSIP) runs for 10 weeks from mid-May to the end of July

- 12 open-source projects chosen with GT and IBM mentors for 2026
- 2-3 students per project
- Weekly training sessions, meetings with mentors, and a final poster session
- VSIP 2025 program hosted ~30 students

Projects included CFD codes, nuclear fusion simulations, container solutions, and open-source AI training


on to Prompt Accessibility, Maintainability, and Rapid Development.

Low Stakes Issues

ʼU Profiling: *Perf* & *Nsys* to highlight bottlenecks.
ıerization: *Apptainer* for MFC with all essential packages/compilers on OSG Access Point.
ıous Benchmarking: Performance improvement of releases.
CI: Source Code Analyzer (*PMD)*, Frontier benchmark case, RMDA MPI cases, etc.

ıitional Boundary Layer Case
Non-Equilibrium Gas Dissociation for 7-Species Model on Flat Plate.
nent: Riemann Solvers to Capture Aerothermodynamic Phenomena.
ıry: Exascale Parallelization Framework (Legion) Vs. Standard Use of OpenMP/OpenACC.
tion: Potential Use of MFC to Design Thermal Protection Systems (TPS).

Student update from MFC project, https://mflowcode.github.io/

MFC was a [Gordon Bell Finalist](#) in 2025!

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Learn more at https://ospo.cc.gatech.edu/vsip/

VSIP Lessons Learned

Successes:

- Most students find the training useful, although it's currently tough to gauge to what extent
  - Certificate programs may help translate this to solid numbers
- Early OSS engagement can lead to continued involvement in OSS projects
  - Several faculty have noted appreciatively that they have enjoyed

Challenges:

- The pairing process remains challenging!
  - 200+ students applied in 2026 for ~26 positions; students rank their top 5 projects according to interest
    - Committee looks to both include students with experience in specific topic areas and engage newer students
  - Using a Google Summer of Code process (students apply directly to a project) may help but requires extra manpower and a longer timeframe

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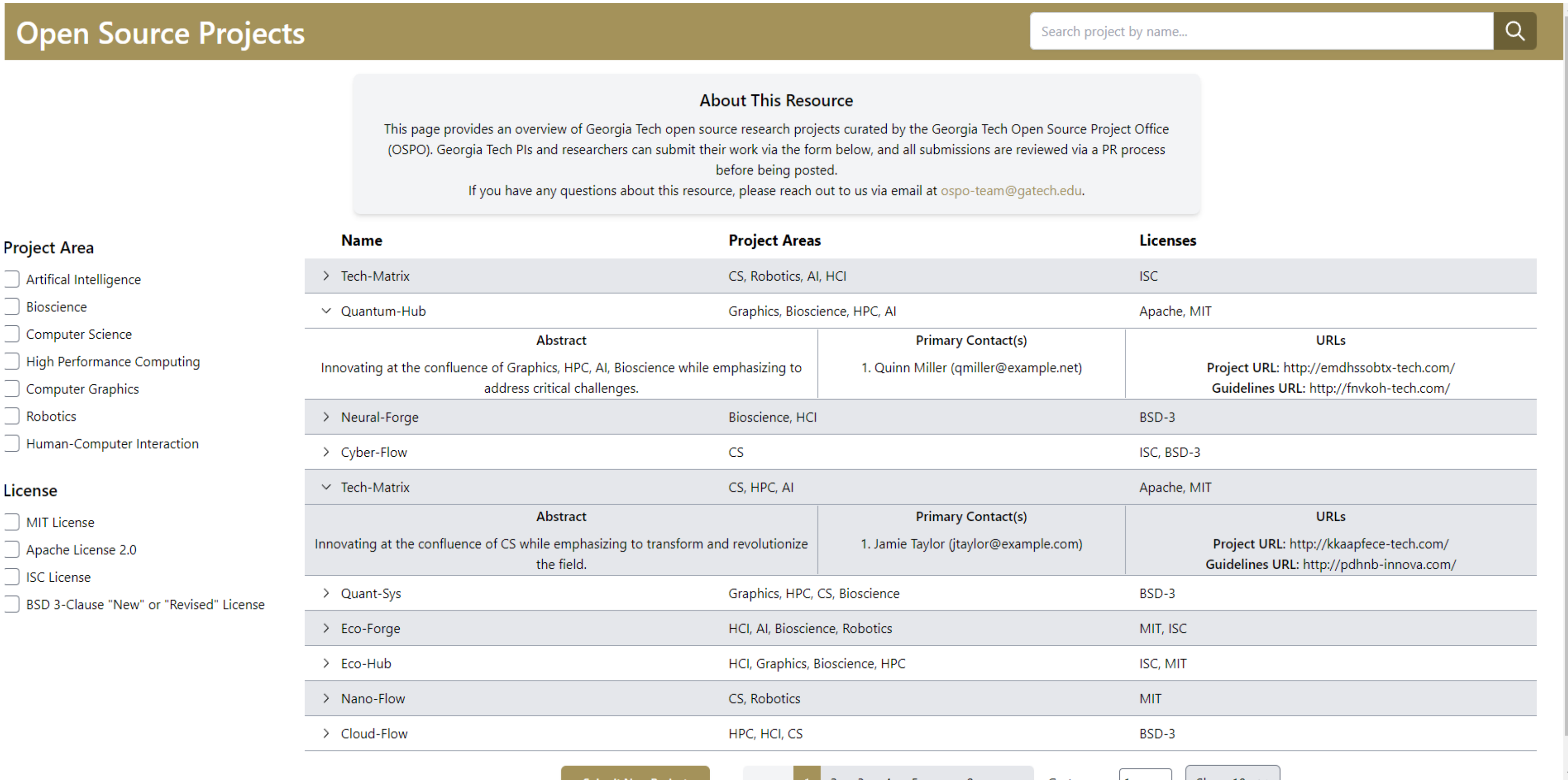
Open Source Project Explorer

A student-built open source GitHub Pages web interface that allows GT researchers:

- To share their open source projects and relevant details
- To explore new projects, licenses, and tools
  - Allows faculty to make new connections with their research
  - Enables students to learn about and contribute to open source projects
- Helps the OSPO to do ***community discovery***



Development by Nathaniel Kim, Kate Rachwal
https://gt-ospo.github.io/oss-project-explorer/

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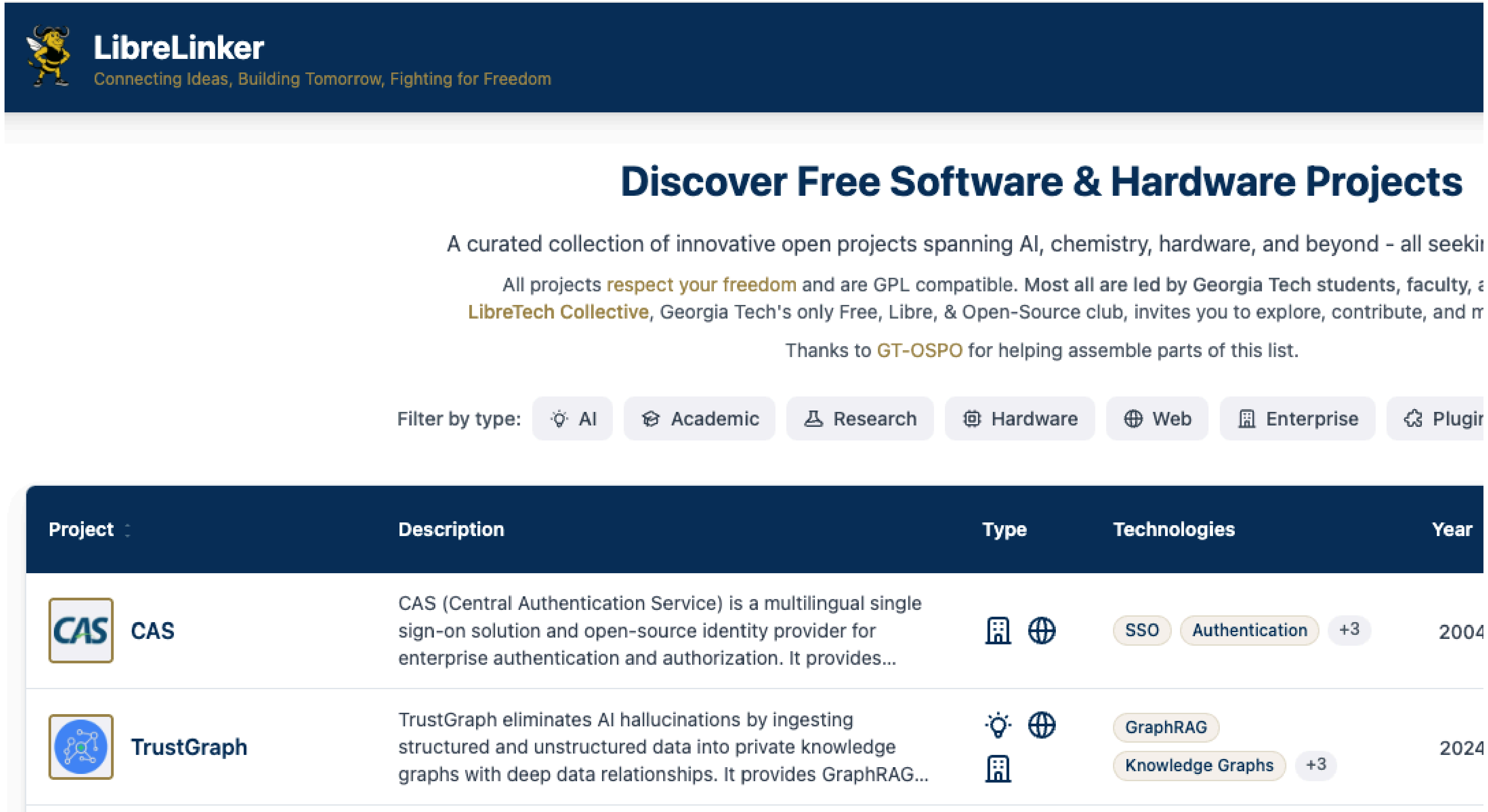
LibreLinker

LTC@GT has a similar tool at https://librelinker.us/ that focuses on projects students can engage with.

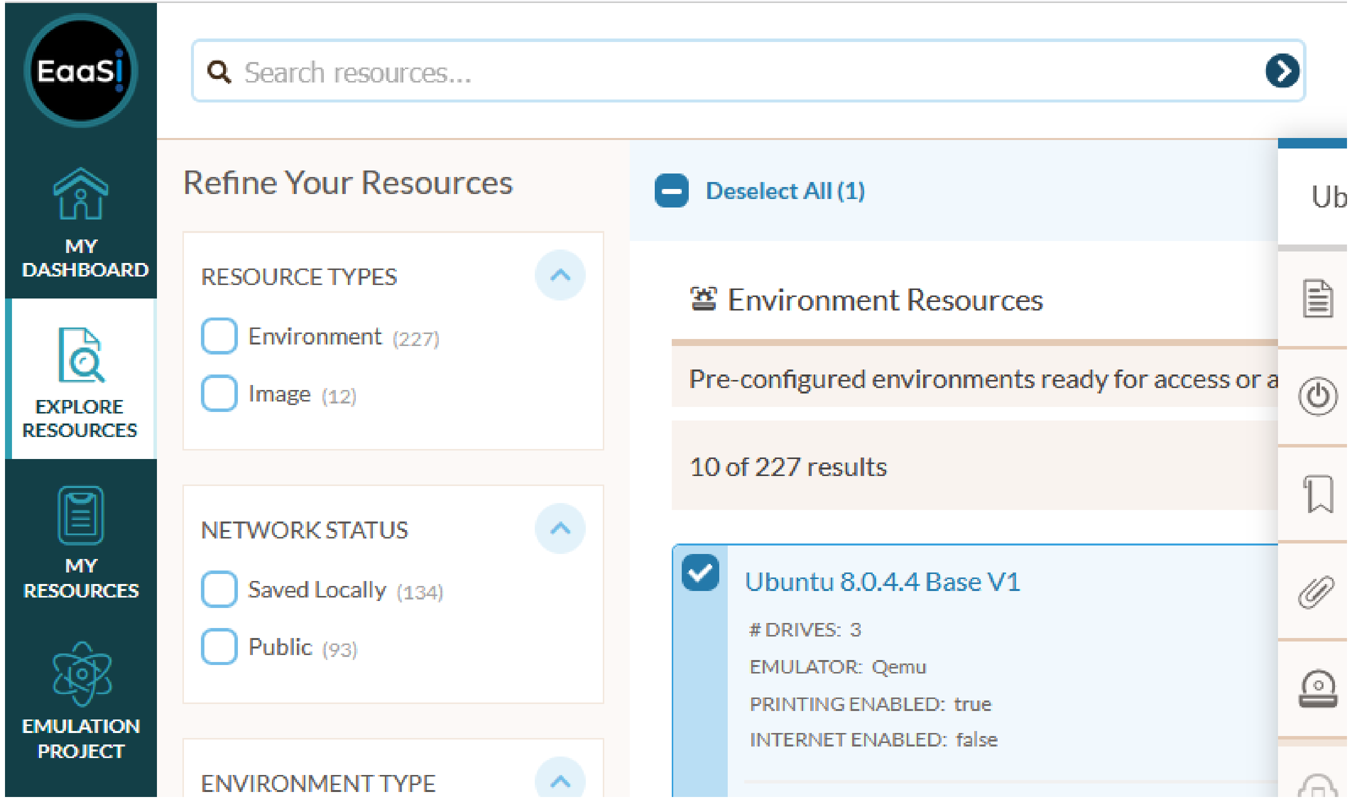


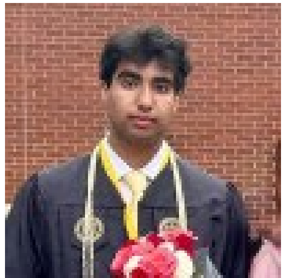
Student Work on Software Preservation

Working closely with the GT Library on exploring the EaaSI platform – Collaboration between Yale and the Software Preservation Network (SPN)

- Also investigated Olive from CMU
- E.g. We may want to build an older version of OpenMPI + Linux Centos 7 workflow for running legacy versions of codes

- Initial codebase released, but technical efforts must also be combined with policy discussions
  - *What is worth the effort of archiving?*





Development by Saad Ata and Ken Shibata
https://github.com/gt-ospo/oss-software-preservation

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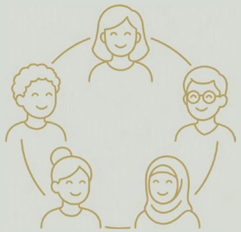
OSPO@GT – Community Building

OSS Community Building



Industry sponsored events to support open-source topics. Poster sessions to share OSS software on GT's campus. Support for student-run clubs

OSPO@GT – Community Building

The OSPO supports different training and community building events throughout the year including:

- Open-source AI workshops with collaborators at IBM
  - Focusing on open source AI projects like Docling, Granite, and Mellea
  - Including student talks and lightning talks on their work with VSIP and internships
- Monthly meetups open to all faculty, students, staff
- Poster sessions for OSS software with our data science institute, IDEaS
- Indirect support for student efforts like LibreTech Collective!

Learn more at https://ospo.cc.gatech.edu/events

OSPO@GT – Open-Source AI Resources

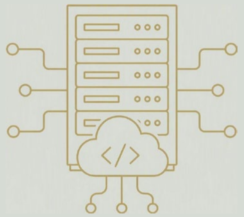
Focus on supporting open-source models and infrastructure

- Complements ongoing vendor-driven efforts with GT’s Office of IT and Library

Support of academic-related consortiums like Open Forum for AI (OFAI) and the OSI open source AI definition

- Participation in working groups around educational front-ends like the Dietrich Analysis Research Education (DARE) platform

Open-Source AI Resources

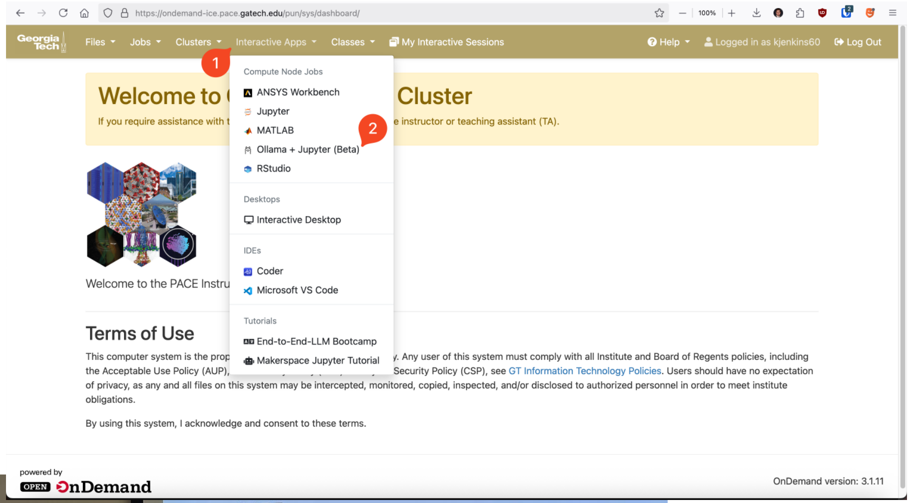


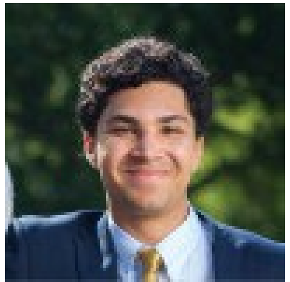
Access to open compute, model hosting, and infrastructure to help students experiment with and deploy open-source AI.

Open-Source AI Resources

We have supplemented paid tools with the following:

- Training and demos for open-source RAG pipelines using open weights models and tools like LangChain
- Ollama+Jupyter workflows for our local HPC resources that focus on the open Granite models
- An in-house, local inference service, SPIN, which utilizes open source AI models
- Explorations of proof-of-concepts for tools like CMU's DARE platform





Development by Dahirou Harden, among other students
Learn more at https://github.com/gt-ospo/granite-workshop
DARE: https://www.cmu.edu/dietrich/ai/education/dare.html

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LibreTech Collective

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LTC@GT Origin Story

- LibreTech was founded in 2025 by Alex Jenkins, then an GT's OSPO VSIP as a student developer.
  - The club is built solely on Free Software principles and computing freedom - for students, by students. Every project must let users Run, Study, Modify, and Share the code; we have no preference on copyleft licenses.
- We chose "free" over "open" - as in freedom and price - because it resonates more than with students than another corporate-branded 'open-source' campus event.
- That same year, LTC was confirmed as the first club of its kind in the world by the FSF. In 2026, it became a Departmental RSO within the College of Computing at GT - the same department as GT's OSPO - making us an independent student-run unit.
- LTC runs on four pillars:
  - Hackathons
  - Guest speaker events
  - Homebrews (on-campus free-software meetups)
  - Student-led free software initiatives

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LTC@GT – A Student Run FLOSS Club!

- Learn more about us at:
  - https://ltc.cc.gatech.edu/ (Home Site)
  - https://librelinker.us/ (GT Free Software Registry)
  - https://codeberg.org/LTC-GT (Active Club Projects)
- Fun Facts:
  - All club materials approved by the FSF for Federation
  - LTC has been featured in coverage from:
    - Hacker news
    - Slashdot
    - It's FOSS
    - FOSS Force
    - Techrights
    - Tuxmachines
    - FSF/GNU Press Publications
    - The Technique (GT Campus Newspaper)
    - And many others! (do a web-search of us!)

GEORGIA TECH
LIBRETECH COLLECTIVE
It’s GNU for you - Hack. Share. Learn.

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LibreTech Opportunities for Engagement

- We are always looking for sponsors for on-campus events! See ltc.gatech.edu/donate
- Any Georgia Tech alumni or student in good standing with Georiga Tech can join LibreTech Collective - submit projects and collaborate with fellow members.
- Over a dozen students have committed directly to projects like GNU and TrustGraph, plus many other on-campus initiatives - we connect students with developers beforehand to give them a head start.

Connect with us!

- Via our websites at ospo.cc.gatech.edu and ltc.cc.gatech.edu
- Discord (student-focused)
- LinkedIn and BlueSky accounts

OSPO@GT Discord
https://discord.gg/x73GXzPPH6

LibreTech Collective Discord
https://discord.com/invite/E6qgerDpTr